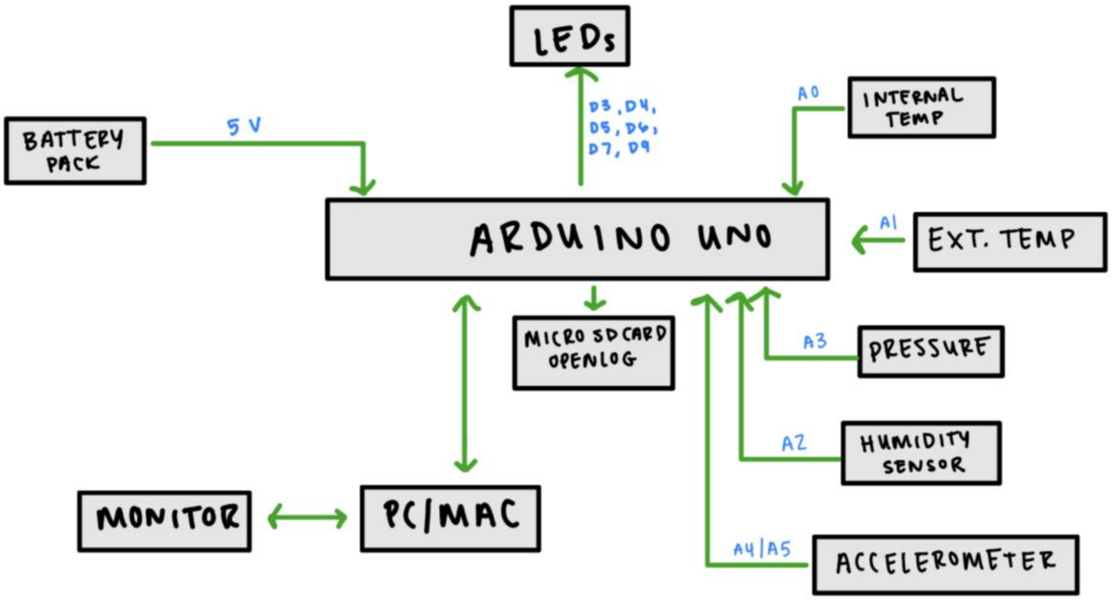
# Balloon Shield Design

## PHASE 1:

### System Block Diagram




### Total Approximate Power Consumption

COMPONENTS AND POWER:

| COMPONENTS | CURRENT |
|---|---|
| Arduino UNO | 50 mA |
| Humidity sensor | 2 mA |
| Pressure sensor | 7 mA |
| Accelerometer | 0.35 mA |
| Temp sensors (x2) | 1 mA (x2) |
| LEDs (x6) | 10 mA (x6) |
| Openlog SD card Module | ≈ 30 mA |
| TOTAL | 151 mA |
| with 5V power supply: | P = I × V<br>P = 151 mA × 5 V<br>$\boxed{P = 0.76\ W}$ |

The total power consumption is estimated to be around **0.76 W** for all onboard components including the OpenLog SD card reader, as it can significantly increase current draw when logging data.

## Total System Weight

TOTAL SYSTEM WEIGHT:

| COMPONENTS | APPROX. WEIGHT |
|---|---|
| Arduino UNO | 25 g |
| Humidity sensor | 3 g |
| Pressure sensor | 5 g |
| Accelerometer | 2 g |
| Temp sensors + wiring | ≈ 5 g |
| LEDs / Resistors | 3 g |
| Balloon Shield | ≈ 20 g |
| OpenLog / SD card Module | 3 g |
| TOTAL APPROX. WEIGHT: | 65g – 70g |

The total system weight with all system components should be around **65-70 g** for an approximation.

## System Feasibility

With certain components being unavailable for purchase, such as the humidity sensor, the system may need to be restructured. One alternative is to purchase a humidity/temp sensor **(Adafruit AM2320)** that can measure both the ambient humidity and external temperature that also has a standard I2C protocol. With this adjustment, however, the PCB shield will need to be redesigned and printed.

Depending on what the maximum altitude is for this balloon design, the operating temperature for each component needs to be taken into account. Most of the electronic components operate on the low end at -40ºC.

Overall, the system can accurately read and log temperature, humidity, pressure, and acceleration data within a small tolerance. Some important things to consider are temperature tolerances, components fragility, and onboard ADC precision for processing data.